

1st Place Solution for the HANDS@ICCV25 Workshop ARCTIC Challenge GHOST: Gaussian Hand–Object Surface Reconstruction with Geometric Priors

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Abstract

We present *GHOST*, the 1st-place solution to the HANDS@ICCV25 ARCTIC challenge for bimanual hand–object reconstruction. *GHOST* reconstructs animatable hand avatars and category-agnostic objects from monocular RGB videos using Gaussian Splatting with realistic hand-object interactions. A geometric prior retrieval stage aligns retrieved 3D models to the scene to recover occluded regions, while a hand-mask-aware background loss prevents penalizing hand-occluded object areas. During optimization, a grasp-aware contact formulation improves the realism of 3D hand–object contact. *GHOST* produces complete, physically consistent reconstructions and achieves state-of-the-art results, outperforming previous methods.

1. Introduction

The ARCTIC track focuses on reconstructing bimanual hand–object interactions from monocular RGB videos in a category-agnostic setup. The task is challenging due to strong mutual occlusions, scale inconsistencies, and missing object regions caused by hand and self-occlusions. *GHOST* addresses these limitations through a Gaussian Splatting framework that combines object geometric priors, hand-aware background losses, and grasp-aware contact optimization. It also integrates temporal filtering to ensure consistent motion across frames. Together, these components produce complete and realistic reconstructions of two-hand-object interactions and achieve 1st place in the HANDS@ICCV25 ARCTIC challenge.

2. Method

The pipeline consists of a preprocessing stage, where object priors and hand reconstructions are obtained and refined, followed by a Gaussian Splatting optimization stage that

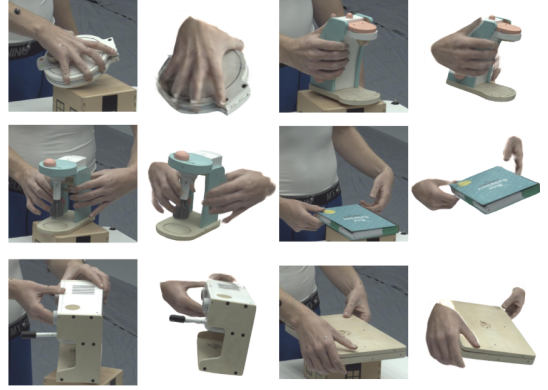


Figure 1. *GHOST* reconstructs 3D Hand-Object interactions from monocular RGB videos using 2D Gaussian Splatting.

jointly reconstructs the hand-object interactions. Figure 2 summarizes the stages of our pipeline. In what follows, we describe each step in detail.

2.1. Preprocessing Pipeline

Let the input video be denoted as $\mathcal{V} = \{I_t\}_{t=1}^T$, with T frames. We begin by segmenting the object of interest using SAM2 [12], producing per-frame masks $\mathcal{M}_t^o$. Using the object’s segmentation, we apply Structure-from-Motion (SfM) to obtain the camera intrinsics $\mathbf{K}$, the relative camera trajectory $\{(\mathbf{R}_t, \mathbf{T}_t)\}_{t=1}^T$, and a sparse object point cloud $\mathcal{P}_{sfm}$ using VGGSfM [14].

2.1.1. Object Geometric Prior

Reconstructing objects under hand manipulation is challenging due to hand and self-occlusions, which often leave unseen regions incomplete. To address this, we leverage open-source large-scale 3D object databases such as Objaverse [3, 4]), which provide extensive coverage of everyday objects. We use these models as geometric priors to refine and complete partial reconstructions, improving both object accuracy and hand-object interaction quality.

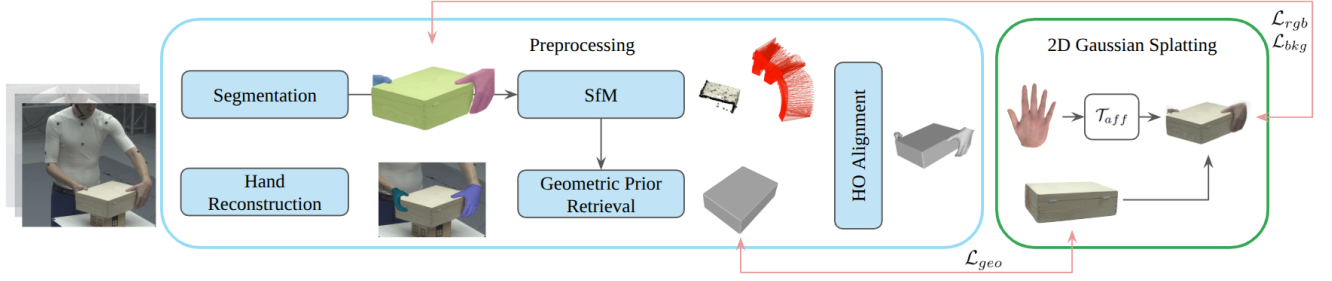


Figure 2. Overview of our pipeline. The preprocessing stage extracts object masks, camera poses, the object’s point cloud, and its geometric prior, while reconstructing and refining hand poses. The Gaussian Splatting stage initializes object Gaussians, rigs hand Gaussians, and jointly optimizes both to reconstruct consistent hand–object interactions in 3D.

Retrieval A textual description d of the hand-held object is obtained from a few sampled video frames using a vision-language model (e.g., InternVL [1, 2]) or provided directly (e.g., ‘Box’). The text embedding $\phi_{txt}(d)$ is computed using OpenShape’s [9] CLIP model, which maps text and object meshes into a shared embedding space. Since all Objaverse objects are pre-embedded, retrieval reduces to a nearest-neighbor search between $\phi_{txt}(d)$ and stored embeddings $\{\phi_{obj}(\mathcal{O}_i)\}$, producing k candidate meshes $\{\mathcal{O}_1, \dots, \mathcal{O}_k\}$.

Prior-Mask Alignment Each retrieved candidate mesh $\mathcal{O}_i$ is then aligned to the scene using the camera poses $(\mathbf{R}_t^c, \mathbf{T}_t^c)$ from SfM and the object masks $\mathcal{M}_t^o$. We optimize a similarity transform consisting of rotation $\mathbf{R}_p$, translation $\mathbf{T}_p$, and scale s_p such that the rendered projection of the candidate best matches the observed masks. The objective is defined as a mask-based IoU loss (i.e., silhouette loss). The candidate with the lowest $\mathcal{L}_{IoU}$ across sampled frames is selected as the object’s geometric prior $\mathcal{O}_p$ for all subsequent optimization stages.

2.1.2. Hand Reconstruction Initialization

We initialize hand reconstructions using an off-the-shelf hand reconstruction model (i.e., HaMeR [11]), guided by joint and bounding-box detections from RTMPose [6–8]. For each frame I_t , this yields MANO parameters consisting of pose θ_t , shape β , global hand rotation $\mathbf{R}_t^h$, and translation $\mathbf{T}_t^h$. To handle jittery or unreliable reconstructions caused by occlusions, we discard low-confidence frames based on detection confidence and deviation from neighboring frames. The removed frames are then interpolated, producing temporally consistent hand vertices $\mathbf{V}_t^h$ and joints $\mathbf{J}_t^h$ across the sequence. In addition, we segment and track the hand masks using SAM2 [12] and crop it using the hand’s 2D keypoints to exclude the forearm, producing $\mathcal{M}_t^h$.

2.2. Hand Translation and Object Scale Optimization (HO Alignment)

Initial hand reconstructions and the object’s point cloud $\mathcal{P}_{sfm}$ may be misaligned in 3D, due to scale inconsistencies

from SfM or drifting hand translations. To correct this, we jointly refine hand translations $\mathbf{T}_t^h$ and global object scale s_o . The optimized scale is then applied to the point cloud $\mathcal{P}_{sfm}$ and camera translations $\mathbf{T}_t^c$.

2.2.1. Grasping Detection

To identify frames where the hands interact with the object, we compare the motion of the object and each hand. The object’s motion $\Delta \mathbf{T}^o$ is computed as the change in its point cloud center across frames. If this motion is below a threshold, the object is considered static. For each hand, we compute its translation change $\Delta \mathbf{T}^h$ and measure the 2D cosine similarity between its motion and object’s, projected to the x, y plane: $c_h = \Delta \hat{\mathbf{T}}_{xy}^o \cdot \Delta \hat{\mathbf{T}}_{xy}^h$. If both hands have $c_h > 0.5$, the frame is labeled as a two-hand grasp; otherwise, the hand with the higher score is considered the interacting one.

2.2.2. Optimization Objective

Once grasping frames are identified, we optimize a joint objective consisting of three complementary loss terms.

- **Contact Loss $\mathcal{L}_{contact}$:** During grasping frames, it encourages contact between the hand mesh vertices $\mathbf{V}_t^h$ and the scaled object geometric prior $\mathcal{O}_p$ using Chamfer distance.
- **Hand 2D Projection Loss $\mathcal{L}_{proj}$:** Penalizes deviations from HaMeR-predicted 2D hand joints, ignoring low-confidence predictions.
- **Temporal Consistency Loss $\mathcal{L}_{temp}$:** Enforces smooth hand translation across time.

2.3. Hand-Object Gaussian Splatting

We represent both the hand and object using sets of 3D Gaussians. The complete Gaussian set $\mathcal{G} = \{\mathcal{G}_o, \mathcal{G}_h\}$ is defined as:

$$\mathcal{G}_o = \{p_o, r_o, s_o, \alpha_o, SH_o\}, \mathcal{G}_h = \{p_h, r_h, s_h, \alpha_h, SH_h\} \quad (1)$$

where p are Gaussian 3D centers initialized by $\mathcal{P}_{sfm}$, r are Gaussian rotations, s are 2D scales of Gaussians, α are opacities, and SH are spherical harmonics that represent the color of the Gaussians.

Table 1. Comparison of 3D metrics across HOLD, BIGS, and our method. All MPJPE metrics are in mm , and all CD metrics are in cm^2 .

Method	MPJPE _{RA,h} ↓	MPJPE _{RA,l} ↓	MPJPE _{RA,r} ↓	CD _h ↓	CD _{ICP} ↓	CD _l ↓	CD _r ↓	F _{10mm} (%)↑	F _{5mm} (%)↑
HOLD [5]	25.91	27.13	24.70	114.73	2.07	105.92	123.54	63.92	37.13
BIGS [10]	35.14	34.17	36.10	38.69	1.36	46.11	31.28	81.78	56.41
Ours	24.07	25.42	22.71	18.40	2.26	23.41	13.40	60.88	34.67

2.3.1. Hand Gaussians Rigging

Let the canonical MANO mesh be defined by vertices $\mathbf{V}^c \in \mathbb{R}^{778 \times 3}$ and triangular faces $\mathbf{F}^c \in \mathbb{R}^{1538 \times 3}$. For each face $f \in \mathbf{F}^c$, we attach m Gaussians parameterized by canonical centers $p_i^c \in \mathbb{R}^3$, rotations $r_i^c \in SO(3)$, and initial scales $s_i^c \in \mathbb{R}^2$, with $i \in \{1, \dots, m\}$.

Given the sequence of deformed meshes $\mathbf{V}_t^h$, we compute per-face affine transformations by constructing basis matrices for the face in canonical and deformed states. The affine mapping from canonical to deformed coordinates is defined as $\mathbf{M}_{f,t} = \mathbf{T}_{f,t}^d \mathbf{T}_f^{c^{-1}}$. Each Gaussian rigged to face f deforms according to

$$p_i^d = \mathbf{M}_{f,t} p_i^c, \quad r_i^d = \text{Polar}(\mathbf{M}_{f,t}), \quad (2)$$

where $\text{Polar}(\cdot)$ extracts the rotational component from the polar decomposition of $\mathbf{M}_{f,t}$. This operation is shown in Figure 2 as $\mathcal{T}_{aff}$.

2.3.2. Optimization Objective

The final optimization subsequently refines object and hand Gaussians in two stages (S1) and (S2) to achieve photometric fidelity, geometric consistency, and occlusion robustness, ensuring coherent rendering of complex hand-object interactions. The optimization is guided by several complementary loss terms:

- Photometric loss $\mathcal{L}_{rgb}$: combines L1 and SSIM losses between rendered images and ground-truth segmentations (S1,2).
- Background loss $\mathcal{L}_{bkg}$ [13]: penalizes Gaussians that project to the background (S2).
- Hand-aware Background loss $\mathcal{L}_{bkg,h}$: standard $\mathcal{L}_{bkg}$ may wrongly penalize object regions occluded by the hand. We therefore exclude hand pixels from the background mask and apply $\mathcal{L}_{bkg,h}$ only to true background regions, conditioned on the hand side h (S1).
- Geometric Consistency loss $\mathcal{L}_{geo}$: enforces alignment between object’s Gaussian centers p_o with the geometric prior $\mathcal{O}_p$: $\mathcal{L}_{geo} = \lambda_{geo}(\mathcal{L}_{out} + \mathcal{L}_{fill})$, where $\mathcal{L}_{out}$ penalizes Gaussians deviating beyond a threshold τ_{out} from the prior surface, and $\mathcal{L}_{fill}$ encourages completion of occluded regions by penalizing distances exceeding τ_{fill} (S1).

3. Experiments

In Table 2, we evaluate the impact of different SfM methods and the geometric consistency loss on object reconstruction

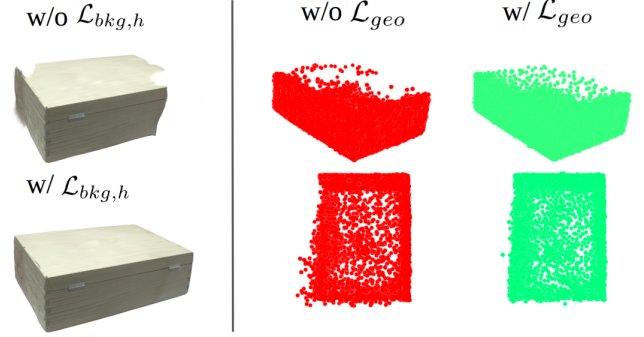


Figure 3. Qualitative comparison for the impact of $\mathcal{L}_{bkg,h}$ (left) and the quality of object’s point clouds obtained by applying $\mathcal{L}_{geo}$ (right).

metrics and interaction metrics. Figure 3 illustrates the effect of excluding hand masks from the background during training and applying $\mathcal{L}_{bkg,h}$, as well as the impact of applying $\mathcal{L}_{geo}$ on the point cloud obtained from Gaussian centers. More qualitative results are shown in Figure 1. Our approach shows consistent improvement in hand reconstruction metrics (MPJPE) and interaction metrics CD_h , CD_r , and CD_l compared to previous methods, see Table 1.

Table 2. Ablation study on the impact of SfM method and geometric consistency loss on object reconstruction and interaction scores. All CD metrics are in cm^2 .

SfM	$\mathcal{L}_{geo}$	τ_{fill}	τ_{out}	CD _h ↓	CD _{ICP} ↓
HLoc	✗	-	-	-	8.18
HLoc	✓	-	-	-	4.86
VGGSfM	✗	-	-	18.32	2.78
VGGSfM	✓	1.0	5.0	18.21	3.66
VGGSfM	✓	0.5	3.0	18.36	2.91
VGGSfM	✓	0.1	2.0	18.42	2.33
VGGSfM	✓	0.01	7.0	18.4	2.26

4. Conclusion

We introduced GHOST, a Gaussian-based framework for reconstructing bimanual hand-object interactions from monocular videos. The method combines geometric priors, hand-aware background losses, and grasp-aware contact optimization to handle occlusions and improve physical realism. GHOST achieves the best performance in the HANDS@ICCV25 ARCTIC challenge, setting a new state of the art for category-agnostic hand-object reconstruction.

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